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HySetter: A Python Package for Reproducible Hydroclimate Data Subsetting over CONUS

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Software

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Summary

Hydrological modeling and research increasingly requires integrating diverse hydroclimate datasets from various sources, which creates a significant technical burden for researchers. This data acquisition process frequently becomes a major bottleneck, diverting valuable time and resources away from scientific analysis. HySetter addresses this challenge by providing a command-line interface for automated and reproducible subsetting of hydroclimate data across the conterminous United States (CONUS). Built upon the HyRiver software stack, HySetter streamlines data acquisition through a unified configuration file that eliminates the need for writing complex scripts for different data sources. By centralizing the data acquisition workflow, HySetter enables researchers to focus on scientific questions rather than data preparation, while simultaneously enhancing the reproducibility and transparency of hydrological research.

Statement of Need

Hydrological research increasingly demands integration of diverse datasets spanning multiple spatial and temporal scales. However, the acquisition, processing, and integration of these datasets presents several challenges:

1. Traditional approaches require developing complex scripts for different data sources, leading to fragmented workflows that hinder reproducibility.
2. Researchers often spend more time on data collection than on scientific analysis and interpretation.
3. The technical overhead disproportionately impacts novice researchers with limited programming skills, hampering their ability to conduct hydrological studies.
4. Even experienced researchers frequently develop redundant boilerplate code for data acquisition across different projects.

These challenges collectively undermine the adoption of FAIR (Findable, Accessible, Interoperable, Reusable) principles (Wilkinson et al., 2016) in hydrological research. While tools like HydroMT (Eilander et al., 2023) have addressed similar challenges for model building and analysis, a comparable solution specifically targeting hydroclimate data acquisition for CONUS remains lacking.

HySetter fills this gap by providing a user-friendly framework that requires minimal technical expertise while supporting advanced applications. Through a simple YAML configuration file and command-line interface, HySetter democratizes access to hydroclimate data, enabling researchers of varying technical backgrounds to efficiently obtain consistent, well-documented datasets for their studies.

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Functionality

HySetter employs a streamlined, configuration-based workflow that can be summarized in four steps:

1. The user defines the area of interest (AOI) in a YAML configuration file, which can be based on Hydrologic Unit Codes (HUCs), GAGES-II basins, NHDPlusV2 catchments, or user-defined geometries.
2. The user specifies the desired datasets and their parameters (variables, temporal ranges, spatial resolutions) in the same configuration file.
3. HySetter processes this configuration to access and subset the relevant data through the HyRiver software stack.
4. The data is standardized, processed, and saved to disk in formats ready for analysis or modeling.

This approach accommodates users with different technical backgrounds: novice users can manually create simple configuration files without writing code, while experienced users can programmatically generate configurations for large-scale or complex analyses.

\autoref{fig:config-example} shows a typical configuration file, and \autoref{fig:config-demo} demonstrates HySetter's command-line interface with progress indicators during data acquisition.

```yaml
project:
  name: hymod
  data_dir: data
aoi:
  gagesii_basins: ['01666500', '02137727']
forcing:
  source: gridmet
  start_date: 2000-01-01
  end_date: 2019-12-31
  variables: [pr, pet]
soil:
  source: polaris
  variables: [bd_5, bd_15, bd_30, bd_60, bd_100, bd_200]
streamflow:
  start_date: 2000-01-01
  end_date: 2019-12-31
  frequency: daily
  use_col: gage_id
```

Figure 1: Example YAML configuration file for HySetter specifying project parameters, area of interest, and data sources.abel{fig:config-example}

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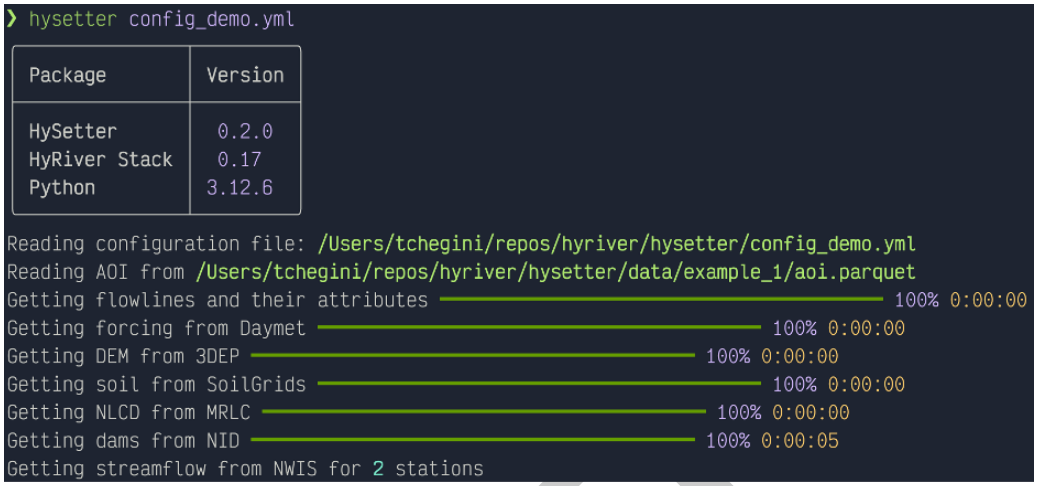

Figure 2: HySetter command-line interface showing real-time progress during data acquisition processes.abel{fig:config-demo}

HySetter currently supports subsetting from the following data sources:

- **Area of Interest**: HUC watersheds, GAGES-II basins, NHDPlusV2 catchments and their attributes (StreamCat and NLDI), or user-defined geometries
- **Drainage Network**: NHDPlusV2
- **Climate Forcing**: Daymet, GridMET, NLDAS2
- **Streamflow**: National Water Information System (NWIS)
- **Soil**: gNATSGO, SoilGrids
- **Topography**: 3D Elevation Program (3DEP)
- **Dams**: National Inventory of Dams (NID)
- **Land Use/Land Cover**: Multi-Resolution Land Characteristics (MRLC) products including land cover, canopy, imperviousness, and urban descriptors

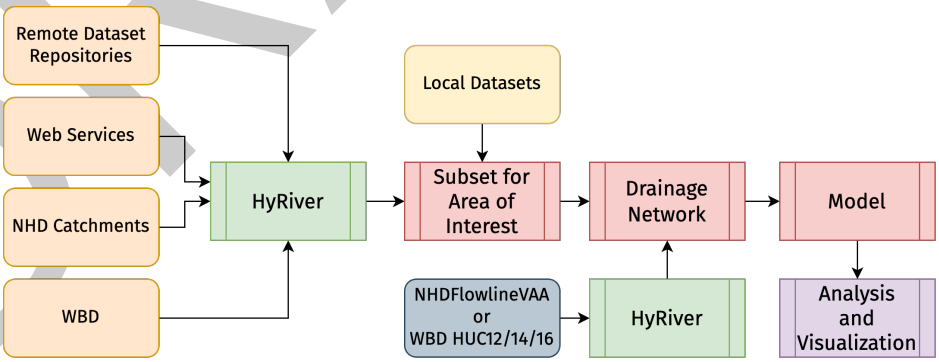

Figure 3: HySetter workflow showing data flow from remote dataset repositories and web services through HyRiver to create area-specific datasets for hydrological modeling.abel{fig:flowchart}

Implementation

The unified workflow of HySetter is illustrated in \autoref{fig:flowchart}, showing how data flows from various sources through the HyRiver stack to create subsets for specific areas of interest. HySetter builds upon the HyRiver software stack (Chegini et al., 2021), which provides access to numerous hydroclimate web services and data repositories. While HyRiver offers powerful capabilities for data acquisition, it requires Python programming knowledge

that can be a barrier for some users. HySetter adds an abstraction layer over HyRiver that translates user-friendly configuration files into the appropriate function calls, enabling access to HyRiver's functionality without writing complex code.

The implementation follows modular design principles with three main components:

1. **Configuration Parser**: Processes YAML configuration files to extract user requirements for areas of interest and data sources.
2. **Data Acquisition Engine**: Interfaces with HyRiver to retrieve and process raw data from multiple sources.
3. **Output Manager**: Standardizes, organizes, and exports the processed data in formats suitable for further analysis.

HySetter provides detailed feedback through progress bars and informative messages during the data acquisition process, helping users track operations and diagnose potential issues. The software includes extensive error handling to gracefully manage common challenges in accessing remote data services, such as connection timeouts or service disruptions.

For advanced users requiring additional post-processing capabilities, HySetter can be used as a Python library with Pythonic and user-friendly API for accessing the downloaded files and further post-processing geospatial operations and integration with other Python libraries.

Research Applications

HySetter has demonstrated its utility in diverse hydrological applications:

1. **Hydrological Modeling**: Implementation and calibration of conceptual and physical models across multiple watersheds, leveraging HySetter's ability to efficiently obtain consistent input datasets for model parameterization. The documentation website provides examples of using HySetter for running simulations using two hydrological models for select watersheds in the US.

2. **Large-Scale Studies**: Analysis of regional hydrological patterns requiring consistent data inputs across numerous basins, benefiting from HySetter's standardized approaches to data acquisition and processing.

3. **Educational Contexts**: Supporting classroom exercises where students can focus on understanding hydrological processes rather than struggling with technical aspects of data acquisition.

The standardized, reproducible nature of HySetter workflows ensures that research outputs can be easily validated and extended by others, promoting transparency and collaboration in hydrological sciences. By abstracting away the complexities of data retrieval, HySetter enables researchers to allocate more time to answering critical hydrological questions and less time to data preparation.

Comparison with Existing Tools

Several tools have been developed to address challenges in hydrological data access and model setup. HydroMT (Eilander et al., 2023) provides a framework for automated and reproducible model building and analysis, focusing on transforming data into model instances. Similarly, SWAT+AW (Chawanda et al., 2020) offers scripted automated workflows for the SWAT model. However, these tools primarily target model building rather than the upstream challenge of standardized data acquisition.

HySetter differentiates itself by focusing specifically on the data acquisition stage, offering a streamlined solution for accessing and subsetting hydroclimate data across CONUS. Unlike more general-purpose tools, HySetter is tailored to the specific datasets commonly used in hydrological

studies over CONUS, providing optimized workflows for these data sources. Additionally, HySetter's YAML-based configuration approach requires minimal technical expertise, making it accessible to users with limited programming experience.

Conclusions and Future Work

HySetter addresses a critical gap in the hydrological modeling workflow by providing a user-friendly, reproducible framework for hydroclimate data acquisition across CONUS. By simplifying access to diverse data sources through a unified configuration-based approach, HySetter enables researchers to focus on scientific questions rather than technical challenges in data preparation.

Future development plans include:

1. Support for additional data sources based on community input and needs.
2. Development of enhanced tools for large-scale studies requiring data acquisition over extensive geographical areas.
3. Creation of additional documentation examples demonstrating HySetter's application in diverse hydrological contexts.

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